AI-Powered Fake News Detection Using Text Classification

Abstract

This project implements a complete machine learning pipeline to classify news articles as real or fake. It evaluates KNN, Logistic Regression, Random Forest, and Neural Networks.

1. Introduction

Fake news poses a significant threat to information integrity.

2. Dataset Description

The dataset used is the Kaggle Fake News dataset.

- Total Articles: 16604
- Fake News Articles: 7661
- Real News Articles: 8943
- Average Words per Article: 307.26
- Missing Text Rows Removed: 0

3. Methodology

Preprocessing included lowercase conversion, punctuation removal, stopword removal, tokenization, and lemmatization.

4. Results

Model	CV Accuracy	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.7062	0.7061	0.6880	0.6641	0.6758	0.7882
KNN	0.5283	0.5347	0.4977	0.9617	0.6560	0.6163
Random Forest	0.6848	0.6889	0.8610	0.3882	0.5351	0.7803
Neural Network	0.6581	0.6744	0.6404	0.6710	0.6553	0.7486
DistilBERT (Fine-Tuned)		0.6945	0.7449	0.5135	0.6079	0.7847

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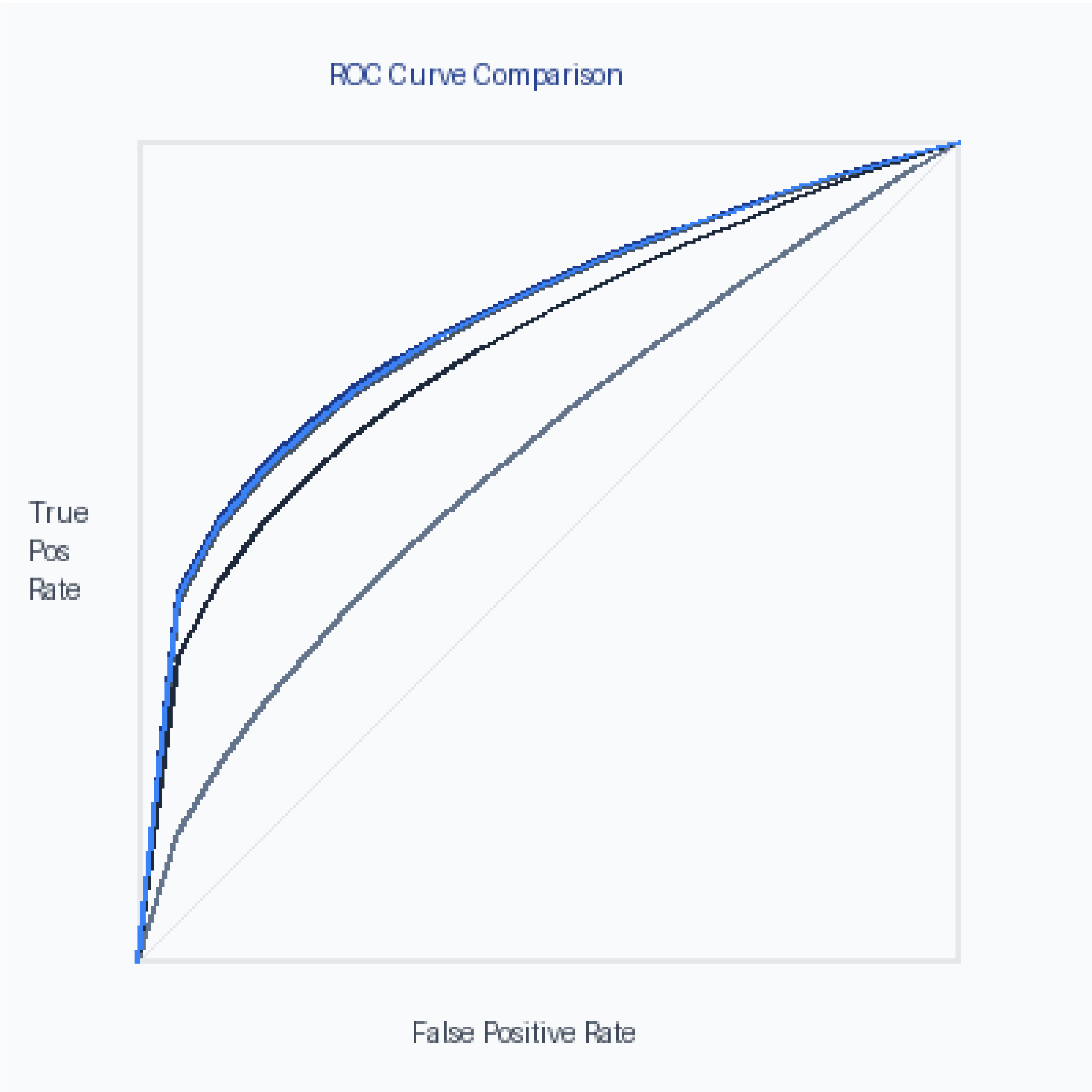

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Confusion Matrices

	Predicted	
	Real	Fake
Real	229	1113
Fake	44	1104

	Predicted	
	Real	Fake
Real	997	345
Fake	386	762

	Predicted	
	Real	Fake
Real	1271	71
Fake	703	445

	Predicted	
	Real	Fake
Real	910	432
Fake	378	770

	Predicted	
	Real	Fake
Real	1141	201
Fake	559	589

5. Discussion & 6. Conclusion

Parametric models like Logistic Regression perform robustly on TF-IDF. Future scope includes deep learning embeddings.